# Sigmoid Function and its Basic Properties

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**Definition:** The sigmoid (logistic) function is given by:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

**Key Properties:**

- Range: $0 < \sigma(z) < 1 \quad \forall z \in \mathbb{R}$
- Symmetry: $\sigma(-z) = 1 - \sigma(z)$
- Derivative: $\frac{d}{dz}\sigma(z) = \sigma(z)(1 - \sigma(z))$

**Sigmoid Function:**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

**1. Symmetry:** Show that $\sigma(-z) = 1 - \sigma(z)$

$$\sigma(-z) = \frac{1}{1 + e^{z}}, \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$1 - \sigma(z) = 1 - \frac{1}{1 + e^{-z}} = \frac{e^{-z}}{1 + e^{-z}} = \frac{1}{1 + e^{z}} = \sigma(-z)$$

**2. Inverse:** Solve $p = \sigma(z)$ for z

$$p = \frac{1}{1 + e^{-z}} \Rightarrow 1 + e^{-z} = \frac{1}{p} \Rightarrow e^{-z} = \frac{1 - p}{p}$$

$$\Rightarrow -z = \log\left(\frac{1 - p}{p}\right) \Rightarrow z = \log\left(\frac{p}{1 - p}\right)$$

# Sigmoid Function: Derivative

**Conclusion:** The inverse of the sigmoid function is the logit function:

$$\sigma^{-1}(p) = \log\left(\frac{p}{1-p}\right)$$

**Proof of Derivative:**

$$\sigma(z) = \frac{1}{1+e^{-z}} \Rightarrow \frac{d\sigma}{dz} = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \left(\frac{1}{1+e^{-z}}\right)\left(1 - \frac{1}{1+e^{-z}}\right) = \sigma(z)(1-\sigma(z))$$

# Introduction to Logistic Regression

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# Introduction to Logistic Regression

**Purpose:** Logistic regression is used to model the probability of a binary outcome (e.g., success/failure, yes/no, 0/1) based on one or more predictor variables.

**Model Form:**

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)}}$$

**Log-Odds (Logit) Form:**

$$\log\left(\frac{P(Y = 1)}{1 - P(Y = 1)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j$$

**Why Use It?**

- Predicts probability while keeping values in [0, 1]
- Interpretable via odds and log-odds
- Suitable for classification problems

# Why Logistic Regression Has This Form

**Problem:** We want to model a probability $P(Y = 1 \mid X = x)$, but a linear model like $\beta_0 + \beta_1 x$ can take values outside the interval [0, 1].

**Solution:** Use the *logit function*, which maps probabilities to the entire real line:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

**Model:**

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x \quad \Rightarrow \quad p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

**Why this form?**

- Ensures $0 < p < 1$ for all $x$
- Logit is the inverse of the sigmoid function
- Interpretable: coefficients show how log-odds change with $x$

# Meaning of Logistic Coefficient $\beta_j$

**Model:**

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j$$

**Interpretation:**

- Each $\beta_j$ represents the change in **log-odds** of $Y = 1$ for a one-unit increase in $X_j$.
- Exponentiating $\beta_j$ gives the **odds ratio**:

$$\text{Odds Ratio} = e^{\beta_j}$$

- If $\beta_j > 0$, the odds of $Y = 1$ increase with $X_j$.
- If $\beta_j < 0$, the odds decrease.

# Example

Suppose a logistic regression yields:

$$\log\left(\frac{p}{1-p}\right) = -1 + 0.7X$$

**Interpretation:**

- $\beta_1 = 0.7 \Rightarrow$ A 1-unit increase in $X$ increases log-odds by 0.7.
- Odds ratio:
$$e^{0.7} \approx 2.01$$
- So, the odds of $Y = 1$ are about **twice as high** for each additional unit of $X$.

# Expected Value in Logistic Regression

**Goal:** Compute $E(Y \mid X)$ in logistic regression.

**Step 1:**

$$p = P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)}}$$

**Step 2: Use expectation of a Bernoulli variable**

$$E(Y \mid X) = 0 \cdot P(Y = 0 \mid X) + 1 \cdot P(Y = 1 \mid X) = p$$

**Final Result:**

$$\boxed{E(Y \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)}}}$$

# Decision Boundary in Logistic Regression

**Logistic Regression:**

$$P(Y = 1 \mid \mathbf{X}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)}}$$

**Decision Boundary:** Predicted class changes at the threshold where

$$P(Y = 1 \mid \mathbf{X}) = 0.5$$

Start with:

$$\frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)}} = 0.5$$

Solve:

$$1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)} = 2$$

$$e^{-(\beta_0 + \dots)} = 1 \quad \Rightarrow \quad \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j = 0$$

# Conclusion

**Decision Boundary:**

$$\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j = 0$$

- It is a **hyperplane** in the input space.
- Separates predictions where $P(Y = 1) > 0.5$ from those where $P(Y = 1) < 0.5$.
- Equivalent to a linear classifier boundary.

# Step-by-Step Logistic Regression

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# Steps Logistic Regression

**Step 1: Define the Problem Goal:** Predict a binary outcome (e.g., Yes/No, 1/0).
**Reason:** Linear regression can predict values outside the (0,1) range, which are not valid probabilities.
**Step 2: Choose the Logistic Model**

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0+\beta_1 X)}}$$

**Reason:** The sigmoid function maps real values to the (0,1) interval, representing valid probabilities.

**Step 3: Construct the Likelihood Function**

Given data points $(x_i, y_i)$, define:

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}$$

**Reason:** This is the joint probability of the observed binary outcomes, assuming independence.

**Step 4: Take the Log-Likelihood**

$$\ell(\beta) = \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

**Reason:** The log function simplifies products into sums and enhances numerical stability.

# Steps Logistic Regression Cont'd...

**Step 5: Maximize the Log-Likelihood Objective:**

$$\frac{\partial \ell}{\partial \beta_j} = 0 \quad \text{analytically or numerically i.e., use gradient ascent}$$

**Reason:** Find parameter estimates that best fit the data by maximizing the log-likelihood.

**Step 6: Interpret the Coefficients** Each $\beta_j$ represents the change in log-odds of $Y = 1$ for a one-unit change in $X_j$.

$$\text{Odds Ratio} = e^{\beta_j}$$

**Reason:** Understand how predictors influence the likelihood of the positive class.

# Steps Logistic Regression Cont'd...

**Step 7: Predict New Outcomes** For new input $x$, compute:

$$\hat{p} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Classify as 1 if $\hat{p} > 0.5$.
**Reason:** Convert probabilities into binary predictions.

# Logistic Regression: Step-by-Step Example using MLE

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# Intercept-only Logistic Regression

A company sends a promotional SMS to 5 customers. The outcomes are:

| Customer | Purchased (Y) |
|---|---|
| 1 | 1 |
| 2 | 0 |
| 3 | 1 |
| 4 | 1 |
| 5 | 0 |

**Goal:** Estimate the overall probability of purchase using logistic regression with only an intercept.

# Model and Log-Likelihood

**Model:**

$$P(Y = 1) = p = \frac{1}{1 + e^{-\beta_0}}$$

Observed: 3 purchases (successes), 2 non-purchases (failures)
**Log-likelihood:**

$$\begin{aligned} \ell(\beta_0) &= 3 \log(p) + 2 \log(1 - p) \\ &= -5 \log(1 + e^{-\beta_0}) - 2\beta_0 \end{aligned}$$

# Maximizing the Log-Likelihood

**Derivative:**

$$\frac{d\ell}{d\beta_0} = \frac{5e^{-\beta_0}}{1 + e^{-\beta_0}} - 2$$

Set derivative to zero:

$$\frac{5e^{-\beta_0}}{1 + e^{-\beta_0}} = 2$$

Let $u = e^{-\beta_0}$:

$$\begin{aligned}
\frac{5u}{1+u} &= 2 \\
5u &= 2 + 2u \\
3u &= 2 \Rightarrow u = \frac{2}{3} \\
\beta_0 &= -\ln\left(\frac{2}{3}\right) = \ln\left(\frac{3}{2}\right) \approx 0.405
\end{aligned}$$

# Final Result

**MLE Estimate:** $\hat{\beta}_0 \approx 0.405$
**Estimated probability:**

$$\hat{p} = \frac{1}{1 + e^{-0.405}} \approx 0.6$$

**Interpretation:** The model correctly recovers the observed success proportion $\frac{3}{5} = 0.6$ without needing gradient descent.

# Logistic Regression $y = \beta_1 x$

In this example, we will solve for the parameter $\beta_1$ in a logistic regression model using **Maximum Likelihood Estimation (MLE)**.
The logistic regression model is given by:

$$p = \frac{1}{1 + e^{-\beta_1 x}}$$

where $p$ is the probability of the outcome, and $x$ is the predictor variable.
We will use the following data:

| Observation | $x$ | $y$ |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 2 | 0 |

Our aim is to estimate $\beta_1$ using the MLE approach.

# Log-Likelihood Function

The log-likelihood function for logistic regression is given by:

$$\ell(\beta_1) = \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Where:

$$p_i = \frac{1}{1 + e^{-\beta_1 x_i}}$$

Substitute the given data $x_1 = 1$, $x_2 = 2$, $y_1 = 1$, and $y_2 = 0$ into the log-likelihood function:

$$\ell(\beta_1) = \log(p_1) + \log(1 - p_2)$$

where:

$$p_1 = \frac{1}{1 + e^{-\beta_1}}, \quad p_2 = \frac{1}{1 + e^{-2\beta_1}}$$

This simplifies to:

$$\ell(\beta_1) = \log\left(\frac{1}{1 + e^{-\beta_1}}\right) + \log\left(1 - \frac{1}{1 + e^{-2\beta_1}}\right)$$

# Simplified Log-Likelihood Function

Simplify the log-likelihood function:

$$\ell(\beta_1) = -\log(1 + e^{-\beta_1}) + \log\left(\frac{e^{-2\beta_1}}{1 + e^{-2\beta_1}}\right)$$

Now, take the derivative of $\ell(\beta_1)$ with respect to $\beta_1$.

# Derivative of Log-Likelihood Function

The derivative of the log-likelihood function is:

$$\frac{d\ell}{d\beta_1} = \frac{e^{-\beta_1}}{1 + e^{-\beta_1}} - \frac{2e^{-2\beta_1}}{1 + e^{2\beta_1}}$$

Set this equal to zero to find the value of $\beta_1$ that maximizes the likelihood function.

$$\frac{e^{-\beta_1}}{1 + e^{-\beta_1}} = \frac{2e^{-2\beta_1}}{1 + e^{2\beta_1}}$$

# Solving for $\beta_1$

Now, solve the equation:

$$\frac{e^{-\beta_1}}{1 + e^{-\beta_1}} = \frac{2e^{-2\beta_1}}{1 + e^{2\beta_1}}$$

This equation can be solved numerically or analytically. For simplicity, we can solve it using a numerical method like Newton-Raphson, as the equation is complex for closed-form solutions.
However, due to the nature of the problem, numerical optimization methods would generally be used here to estimate $\beta_1$.

# Conclusion

In this example, we have derived the log-likelihood function and its derivative for a simple logistic regression model without an intercept. We showed that the derivative of the log-likelihood function can be set to zero to estimate $\beta_1$.
However, due to the complexity of the equation, solving for $\beta_1$ exactly often requires numerical optimization methods, such as Newton-Raphson, to find the optimal value.

# Scenario: Predicting Customer Purchase

A marketing analyst wants to predict whether a customer will buy a product ($Y = 1$) or not ($Y = 0$) based on the number of email ads they've received ($x$).

**Data (sample of 5 customers):**

| $x$ | $y$ |
|---|---|
| 1 | 0 |
| 2 | 0 |
| 3 | 1 |
| 4 | 1 |
| 5 | 1 |

# Aim and Logistic Regression Model

**Aim:** Estimate the probability that a customer buys a product based on emails received.

**Logistic (sigmoid) model:**

$$P(Y = 1 \mid x) = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad \text{where } z = \beta_0 + \beta_1 x$$

**Regression line:**

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

# Constructing the Likelihood Function

Let $p_i = \sigma(\beta_0 + \beta_1 x_i)$

The likelihood function (joint probability):

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}$$

Taking log to simplify:

$$\log L(\beta_0, \beta_1) = \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

**This is the log-likelihood.**

# Substitute Logistic Function into Log-Likelihood

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

**Log-likelihood becomes:**

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^{n} \left[ y_i \log\left(\frac{1}{1 + e^{-z_i}}\right) + (1 - y_i) \log\left(1 - \frac{1}{1 + e^{-z_i}}\right) \right]$$

$$= \sum_{i=1}^{n} \left[ y_i z_i - \log(1 + e^{z_i}) \right], \quad \text{where } z_i = \beta_0 + \beta_1 x_i$$

# Step-by-Step Maximization

1. **Start with initial guesses for** $\beta_0, \beta_1$ (e.g., 0)
2. **Compute:**

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^{n} \left[ y_i(\beta_0 + \beta_1 x_i) - \log(1 + e^{\beta_0 + \beta_1 x_i}) \right]$$

3. **Use gradient ascent (or software like R, Python) to find:**

$$\frac{\partial \ell}{\partial \beta_0}, \quad \frac{\partial \ell}{\partial \beta_1}$$

4. **Iterate until convergence.**

**Final Output:** Estimated coefficients $\hat{\beta}_0, \hat{\beta}_1$ that maximize the likelihood.

# Interpretation

**Example:** Suppose MLE gives $\hat{\beta}_0 = -3.5, \hat{\beta}_1 = 1.2$
**Predicted probability for a customer with 3 emails:**

$$p = \frac{1}{1 + e^{-(-3.5+1.2\times3)}} = \frac{1}{1 + e^{-0.1}} \approx 0.525$$

**Interpretation:** A customer who received 3 emails has a 52.5% chance of buying the product.